Spatial Degradation of Hepatocyte Zonation

Decoding the collapse of liver cell identity across the fatty-liver disease course

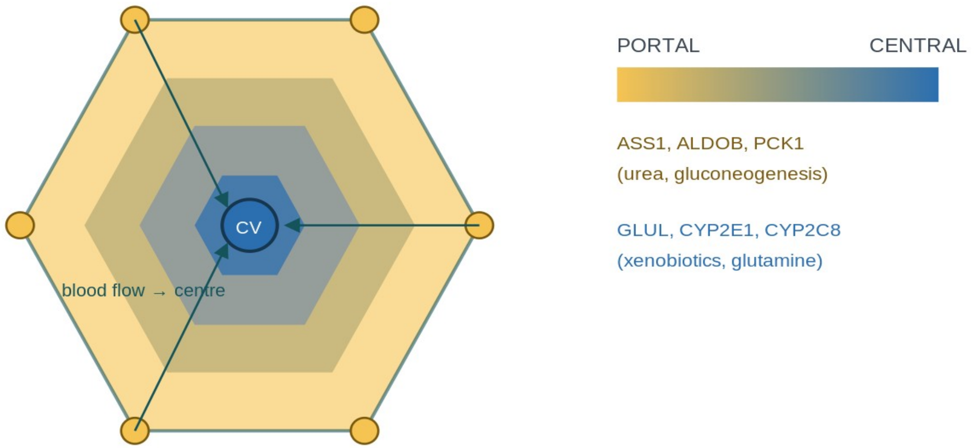
portal central

Roee · Shira | Computational Genomics 76553 — Final Hackathon

What the liver does — and the zonation gradient

The liver is a metabolic factory:

- buffers blood sugar,
- makes plasma proteins
- produces bile acids
- converts nitrogen waste to urea
- detoxifies drugs.



Periportal at the corners; pericentral at the central vein

Two epithelial cells:
hepatocytes (the bulk factory)
and cholangiocytes (bile-duct lining).

Blood flows portal → central, so a cell's environment — and the genes it switches on — depends on where it sits:

- **Periportal:** urea cycle, gluconeogenesis (ASS1, PCK1).
- **Pericentral:** drug metabolism, glutamine (CYP2E1, GLUL).

Zonation = this position-dependent gene gradient.
The hinge: position is encoded in expression, so it can be decoded back out.

WHAT WE MEASURE

Two ways identity breaks: de-zonation vs plasticity

Both look like 'a cell expressing two opposite things at once' — the difference is WHICH boundary is crossed.

De-zonation

Loses its POSITION on the portal ↔ central axis

Co-expresses opposite zonal programs (e.g. CYP2E1 + ASS1).

Signature: the pericentral / periportal zonation genes.

Plasticity

Loses its CELL-TYPE identity (hepatocyte → bile-duct).

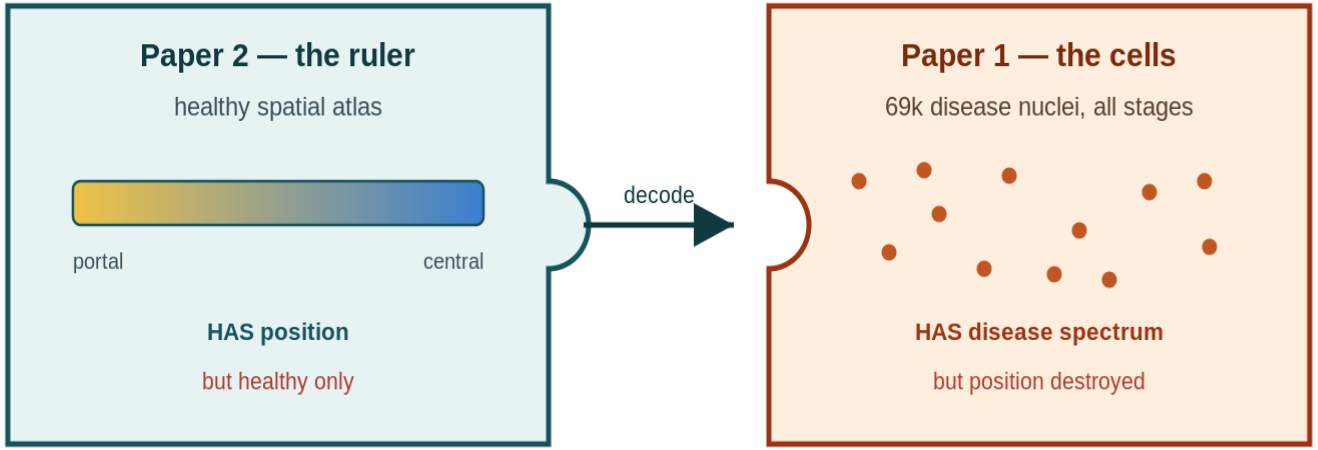
Co-expresses hepatocyte + duct programs (e.g. ALB + KRT7).

Signature: KRT7, KRT19, SOX9 … (from Paper 1).

Analogy: a worker who forgets which station they man (de-zonation) is still a factory worker; one who forgets they work in the factory at all (plasticity) is something else. **H3 asks whether these are the same cells — does losing your address predict losing your job?**

THE OPPORTUNITY

Two datasets, one gap



Paper 2 — the ruler

Healthy human liver with spatial coordinates KEPT
→ a quantitative porto-central zone index.

But healthy only.

Paper 1 — the cells

69k disease nuclei across all stages,
but snRNA-seq THREW AWAY position.

Our move: use the healthy ruler to assign each diseased cell a pseudo-zonal coordinate. Neither paper did this.

WHAT WE TEST

Research question & hypotheses

Q Using the healthy zonation ruler, can we measure how — and how much — hepatocyte zonation degrades as MASLD advances from healthy to end-stage, and is that degradation linked to plasticity?

H1 — Erosion

A de-zonation score rises monotonically Healthy → end-stage.

Test: ordered trend on per-donor values.

H2 — Which programs

Specific programs (e.g. pericentral CYP detox) collapse earliest.

Test: pseudobulk donor×zone DE + FDR.

H3 — Plasticity link

The most de-zonated cells are the ones gaining bile-duct identity.

Test: within-donor correlation.

Why it matters: drug dosing (pericentral CYP) · ammonia/encephalopathy (periportal urea) · finer-than-histology staging · cancer/de-differentiation link · a reusable decoder for spatially-blind atlases.

THE METHOD

How we decode position

1 Signatures

pericentral & periportal gene sets from the healthy atlas

2 Coordinate

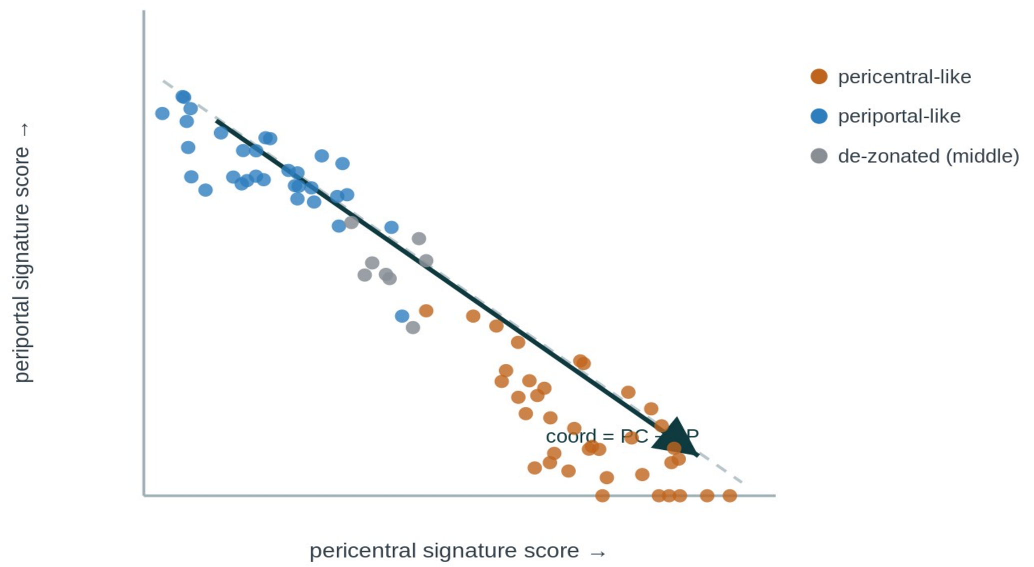
score = mean_z(PC) − mean_z(PP) per cell

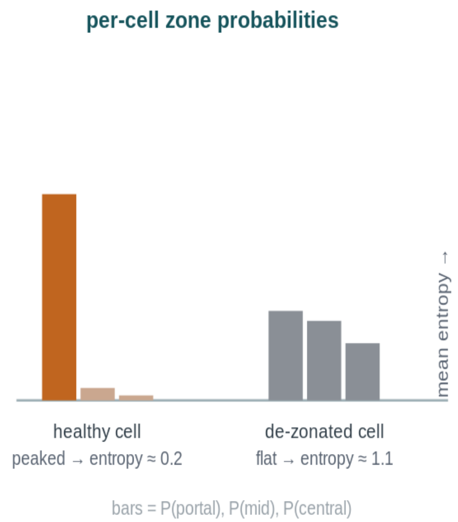
3 Classifier

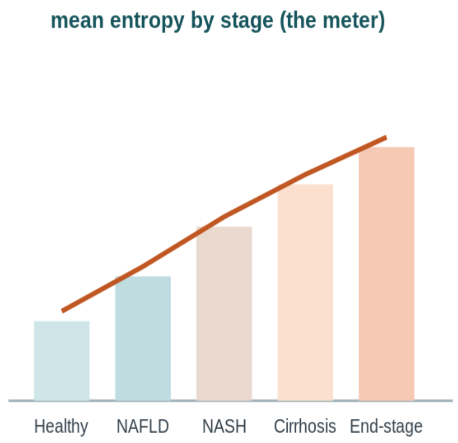
zone probabilities → entropy = confusion = de-zonation

4 Collapse

one metric per donor, tested across stages







WHY IT HOLDS UP

Doing it honestly

Donor-level

Aggregate to one value per donor before testing

cell-level p-values are pseudoreplication and invalid.

No circularity

Genes that build the coordinate are excluded from 'driver' claims; held-out gene split + interaction test for H2.

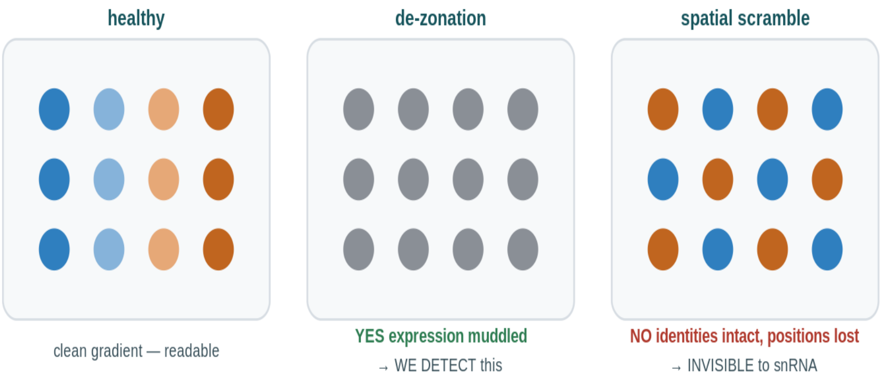
Controls

Positive control (healthy must recover zonation) + negative control (shuffle stage labels → trend vanishes).

What de-zonation is — and isn't:

IS — loss of the expression program (mutual exclusivity weakens, spread narrows).

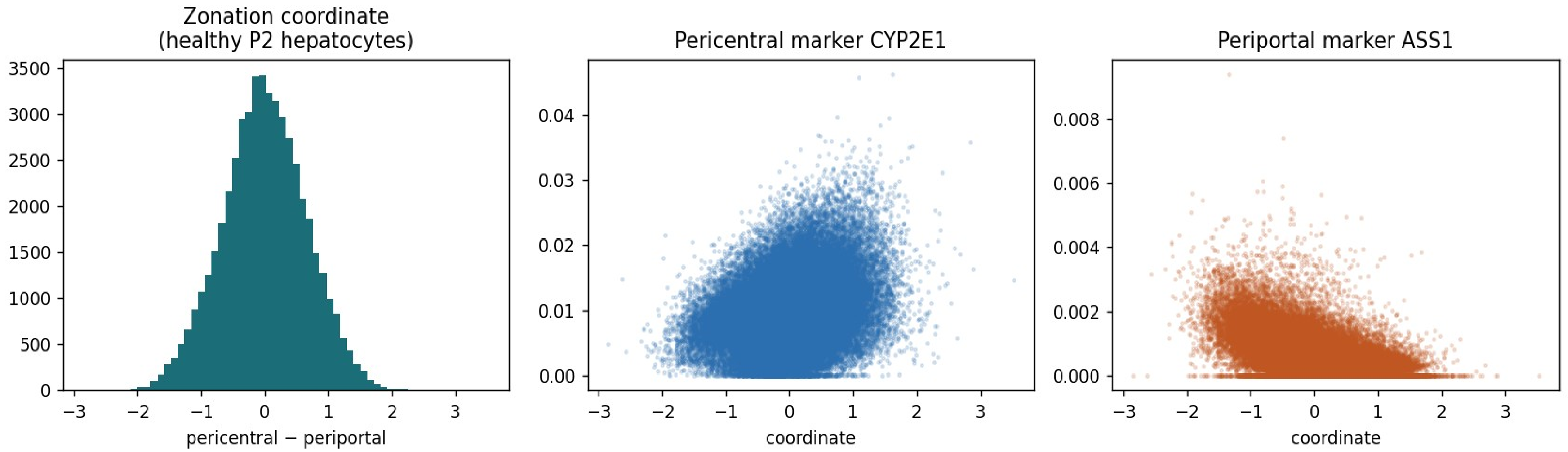
ISN'T — physical spatial scrambling with intact expression; very-unlikely.



POSITIVE CONTROL

✓ DONE — REAL DATA

The ruler works on healthy liver



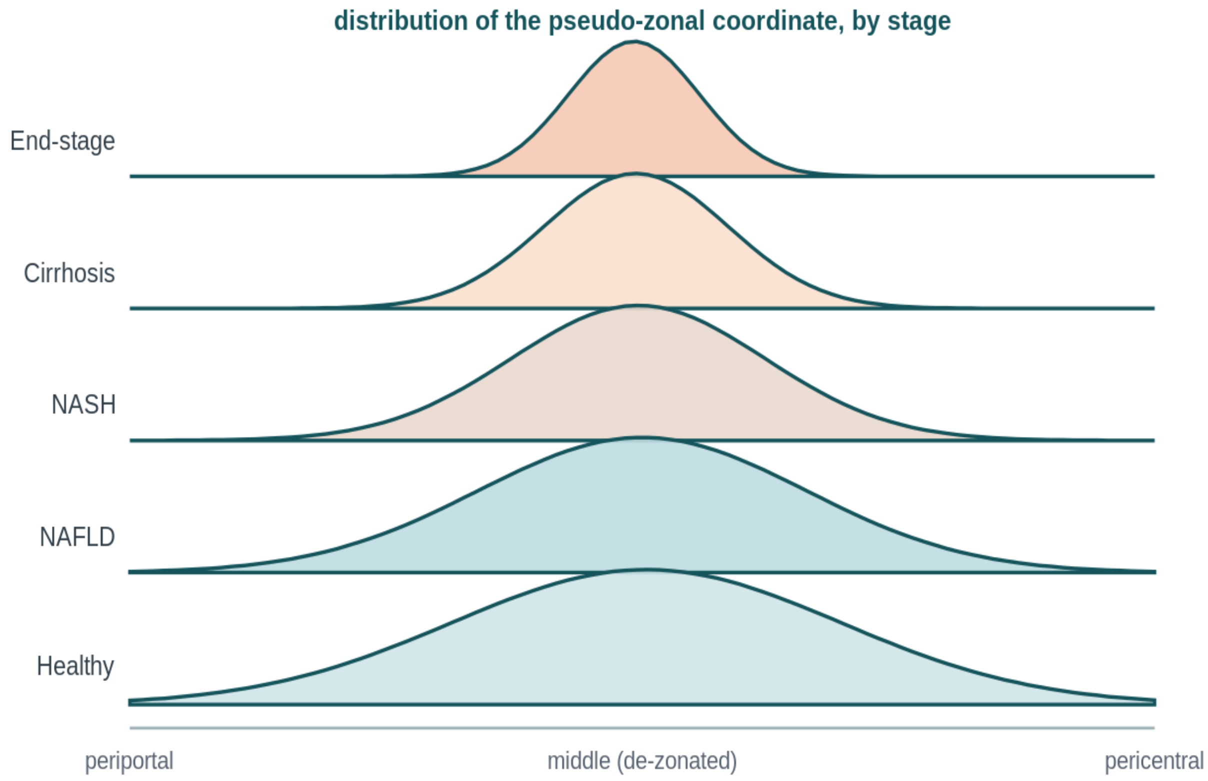
On real Paper 2 healthy hepatocytes the coordinate recovers zonation: pericentral CYP2E1 rises with it and periportal ASS1 falls.

A single broad central hump is exactly what a continuum looks like — the proof of zonation is the marker slopes, not two peaks. We validated before trusting it on disease.

RESULT

H1 — Does zonation collapse across stages?

TO FILL AFTER HACKATHON



EXPECTED SHAPE — swap for real A6

What to look for

Coordinate spread NARROWS and pericentral–periportal anti-correlation weakens toward 0 across stages.

We found: ___

[fill: ρ, donor-bootstrap 95% CI, permutation p]

plot_a6_collapse · pipeline.py:collapse

RESULT

H2 — Which programs collapse, and where?

TO FILL AFTER HACKATHON

FIGURE GOES HERE

Volcano per zone (portal / central): effect vs −log10 q

plot_a7_volcano · signature genes flagged

What to look for

A coherent set of genes at q<0.05 — surviving EXCLUSION of the signature genes — concentrated in one zone first.

We found: ___

[fill: # genes q<0.05 per zone; name the 3–5 earliest-collapsing programs]

pipeline.py:de

RESULT

H3 — Are de-zonated cells the plastic ones?

TO FILL AFTER HACKATHON

FIGURE GOES HERE

Stage-stratified scatter: de-zonation vs plasticity score

plot_a8_plasticity · WITHIN donor/stage

What to look for

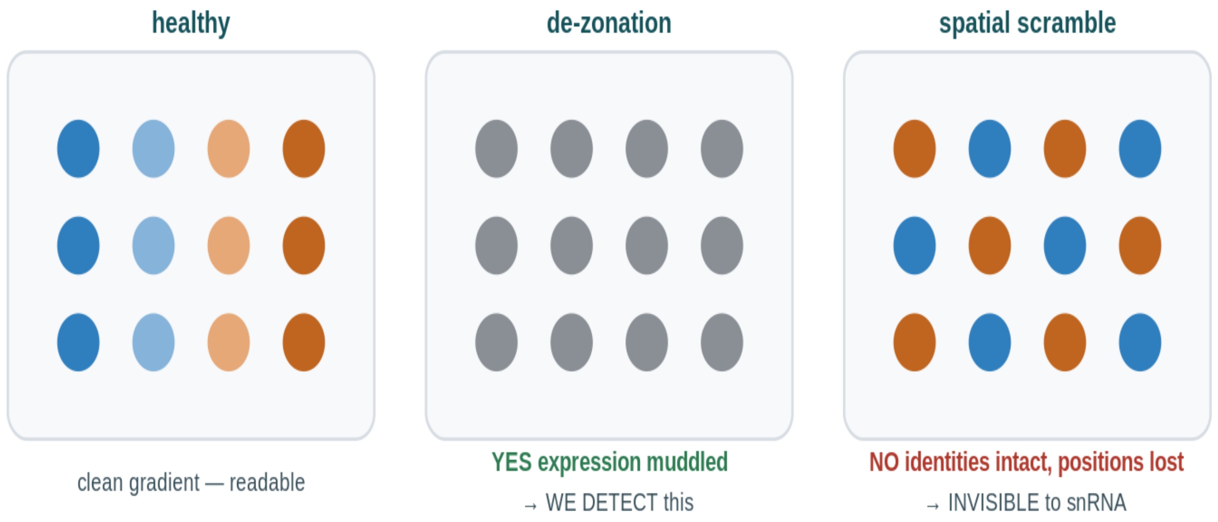
A positive within-donor link — not a pooled one (late disease trivially has more of both).

We found: ___

[fill: mean per-donor ρ; OLS dez coefficient controlling stage+donor]

pipeline.py:plasticity

Limitations & honest scope



- **Blind to pure spatial scrambling** — if cells keep identity but move, dissociated data sees nothing.
- **Pseudo-zonal, not (x,y)** — a 1-D reconstructed coordinate, never a true position.
- **Ruler-validity matters** — disease may corrupt the marker genes; Step 5b checks whether the ruler still holds.
- **Associational** — we measure collapse, not its cause.

TO FILL AFTER HACKATHON

Conclusions

H1 — [did zonation collapse, and how monotonically? ___]

H2 — [which programs eroded first, and in which zone? ___]

H3 — [did de-zonation track plasticity, within donors? ___]

One-line takeaway: a reference-decoder turns the many spatially-blind disease atlases into a quantitative readout of how liver identity disintegrates. [refine with results]

Thank you — questions?

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BACKUP

Is the ruler still a ruler? (validity battery)

TO FILL AFTER HACKATHON

FIGURE GOES HERE

Per-stage: internal coherence · cross-program anti-corr · split-half · program-off vs restriction-lost

plot_a5b_ruler (Step 5b)

BACKUP

Classifier — held-out confusion matrix (Paper 2)

TO FILL AFTER HACKATHON

FIGURE GOES HERE

Confusion matrix + held-out accuracy on Paper 2 before applying to Paper 1

Step 4b · calibrated multinomial logistic regression

Statistical rigor — the checklist

- BH-FDR q<0.05, never raw p
- Ordered trend test, not pairwise only
- Donor-level resampling (never 69k cells as independent)
- Pseudobulk donor×zone DE; H2 via coordinate×stage interaction on held-out genes
- Positive + negative controls
- Effect sizes beside p/q

BACKUP

Bonus — collapse vs inherited risk (Paper 3)

OPTIONAL — IF IT RAN

FIGURE GOES HERE

Hypergeometric overlap: Step-7 collapse drivers vs Paper-3 risk-variant targets, expression-matched background

step9_bonus_enrichment.py · report fold + BH q

Data & reproducibility

Raw → processed: Seurat .rds → counts.mtx; MATLAB .mat → paper2_train.npz. Pipeline reads only data/processed/.

Repo: src/{prep,steps,plotting,tools}, signatures/{paper2_landmark,core,expanded,sensitivity}, results/. run_all.py reproduces every figure on a clean checkout.

Signatures: baseline = EXACT Paper 2 landmark set (paper2_landmark); expanded from the ranked snRNA table; core = curated.